GOVERNANCE, PRIVACY & BUSINESS CONSULTING · LESSON 2 OF 5

Requirements, User Stories, PRDs, Data Storytelling and Agile Delivery for Analytics

Specify analytics work so that it gets built right the first time. Write requirements, user stories and acceptance criteria, draft a product requirements document, communicate findings with data storytelling, and run analytics delivery in short agile cycles with a backlog, sprints and reviews.

Gap-fill lesson · ✦ 2026 · Intermediate · ⏱ 120 min

By the end of this lesson you can

- Write user stories with acceptance criteria that can be tested
- Separate functional and non-functional requirements
- Draft a short product requirements document for an analytics product
- Structure a data story and run agile delivery with a backlog and sprint reviews

Requirements are a contract about meaning

Most failed analytics projects were not technically hard. They built what was asked, and the person asking meant something different. A requirement is a statement of what the solution must do, written so that the builder and the user agree on the meaning. Writing it down is cheaper than rebuilding a dashboard three times.

CASE STUDY **The dashboard nobody used**

A team built a churn dashboard with forty charts. The sponsor wanted to decide, each Monday, which customers to call. The dashboard answered many questions and none of the Monday decision. A short user story, "As the retention lead, I want a Monday list of customers at risk, so that I can call the top 50 before Wednesday", would have shaped a single table and saved six weeks.

User stories and acceptance criteria

A **user story** describes value from the user's point of view:

> As a **[role]**, I want **[need]**, so that **[benefit]**.

The benefit matters most. It tells the builder which details are essential and which are decoration. An **acceptance criterion** says how to test that the story is done, using the form:

> Given **[context]**, when **[action]**, then **[result]**.

Example. As an operations manager, I want a daily list of late orders, so that I can call customers before 10:00.

Acceptance criterion	Test
Given an order is 2 hours or more late, when the list is generated at 07:00, then the order appears on the list	Create a late test order; check the list
Given an order was delivered, when the list is generated, then the order does not appear	Check a delivered order is excluded
Given there are no late orders, when the list is generated, then the list shows "no late orders today"	Run on a day with zero late orders

Each criterion is a test. If a criterion cannot be tested, rewrite it until it can.

Functional and non-functional requirements

Functional requirements say what the solution does. **Non-functional** requirements say how well it does it: speed, accuracy, security, availability, cost, and how long data may be kept.

Type	Example
Functional	The list shows order id, customer, days late and the owner
Performance	The list is ready by 07:00 for 95% of days
Accuracy	Days late matches the logistics system within one hour
Security	Only the operations team can see customer contact details
Availability	The list is available during working hours, 99% of days

Non-functional requirements are often forgotten and then cause the most expensive rework. Write them down at the start.

A short PRD

A **product requirements document (PRD)** gathers the story, the requirements and the success measures for a product, in one or two pages. For an analytics product, a useful PRD has these sections:

1. **Problem and decision.** The problem statement and the decision the product supports.
2. **Users and their stories.** Each user role with its stories and acceptance criteria.
3. **Data.** Sources, grain, freshness and owners.
4. **Requirements.** Functional and non-functional, with priority (must, should, could).
5. **Out of scope.** What is deliberately not included.
6. **Success measures.** The metric and target, and how it will be measured after launch.
7. **Open questions.** What must be answered, by whom and by when.

::::tip The out-of-scope section protects the team Most scope creep arrives as a "small addition". A written out-of-scope list lets the team say "that is a good idea for the next version" with a reason, rather than negotiating it from scratch each time. ::::

Data storytelling

A data story turns findings into a decision. The usual structure is short:

1. **The answer first.** One sentence: "Churn rose four points in March, driven by two plans."
2. **The evidence.** One chart per message, labelled so that it can be read without the speaker.
3. **The decision.** What should happen, who owns it, and by when.

Weak	Stronger
"Here are twelve charts about churn."	"Churn rose four points in March. Fix onboarding for the two plans driving it by 15 May."
A chart with no title and a rainbow palette	A chart titled with its message, one highlighted series
Ending with "any questions?"	Ending with the decision needed and the owner

The same data can support different stories for different audiences. An executive needs the decision; an analyst needs the method. Prepare the first slide for the decision and keep the method in an appendix.

Agile delivery for analytics

Agile delivers work in short cycles, so that the user sees something useful early and changes direction cheaply. The core parts are:

Part	What it is	Analytics example
Backlog	Work ordered by value	"Late order list", "churn drivers", "margin by region"
Sprint	A fixed period, usually one or two weeks, with a goal	Sprint goal: first version of the late order list
Review	Showing the increment to stakeholders	Demo to the operations manager with real data
Retrospective	What to improve in the way the team works	The data delay was found too late; add a freshness check

Kanban is an alternative that shows work moving across columns (to do, in progress, review, done) and limits the number of items in progress. It suits support-style analytics work with unpredictable requests.

Delivery management adds three controls: a scope list that changes only through the backlog, a risk register that names the risks and their owners, and a dependency list that shows which work waits for which data or person.

INTERVIEW CORNER

How do you keep an analytics project from drifting?

Write the problem statement and the user stories first, with acceptance criteria. Keep the scope in a backlog with an out-of-scope list. Deliver a first version within a sprint and review it with the user on real data. Each change goes through the backlog with its benefit stated, so that the trade-off is visible.

::::key What to remember - Write user stories with a benefit and acceptance criteria that can be tested. - Specify non-functional requirements (speed, accuracy, security, availability) early. - A PRD states the decision, the users, the data, the requirements, the out-of-scope list and the success measures. - Storytelling: the answer first, one chart per message, and the decision with an owner. - Deliver in short cycles, review on real data, and change scope through the backlog. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is the most important part of a user story?

A The number of words

B The benefit, which shows what matters most to the user

C The author's name

D The tool to use

Answer: B. The benefit guides what is essential.

2. Which is a valid acceptance criterion?

A The dashboard should look nice

B Given an order is two hours late, when the list runs at 07:00, then the order appears on it

C Users will be happy

D The data should be fast

Answer: B. A criterion has a context, an action and a testable result.

3. Why write non-functional requirements at the start?

A They are required by agile

B They are often forgotten and cause the most expensive rework when discovered late

C They replace functional requirements

D They are optional

Answer: B. Speed, security and accuracy shape the design.

4. In data storytelling, what should come first?

A The method

B The list of charts

C The answer in one sentence

D The thank-you slide

Answer: C. The answer first respects the audience's time.

5. What is the purpose of a sprint review?

A To close the project

B To show the working increment to stakeholders and gather feedback on real data

C To assign blame

D To choose the tool

Answer: B. Reviews make feedback concrete and early.

EXERCISE **Write the spec for one dashboard**

Write a one-page PRD for a dashboard that a sales manager will use each Monday: the decision, two user stories with three acceptance criteria each, four non-functional requirements, the out-of-scope list, two success measures and two open questions. Then write a three-sentence data story for its first review.

Show solution (worked answer)

Hints: make each acceptance criterion testable against the data. Put at least one security requirement in the non-functional list. The data story must lead with the answer, even if the answer is a question the dashboard will settle.

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